Project Overview

This document presents Phase II of our final-year project on the development of a physics-based digital model and intelligent battery state estimations for lithium-ion batteries.

The project aims to establish a modular simulation framework that combines an equivalent-circuit battery model, electrothermal dynamics, battery-pack modelling, sensor measurements, state estimation, protection logic, and degradation-aware control. The framework is being developed in MATLAB and Simulink, with the long-term objective of supporting experimental validation and eventual hardware implementation.

Phase II: Objectives and Planned Work

Phase II focuses on improving the consistency, integration, and validation of the existing model while extending it towards a battery-pack simulink model with intelligent estimation and control.

1. **Complete the mathematical documentation**

   Derive the governing equations for the two-RC electrical model, including both the R_1C_1 and R_2C_2 polarization branches, terminal voltage, SOC, thermal dynamics, charge throughput, capacity degradation, and resistance growth.

2. **Integrate and verify the battery-pack model**

   Verify the series-pack equations, common current, summed terminal voltage, pack resistance, pack temperature, aggregate SOC and SOH, available capacity, and throughput reporting. Reconcile the current eight-cell implementation with the topology of the professor-provided reference model before extending it.

3. **Improve BMS protection and supervisory control**

   Correct the charge/discharge overcurrent logic, verify all fault flags and fault codes, check state transitions, and validate that emergency shutdown forces the safe-current command to zero.

4. **Strengthen SOC and SOH estimation**

   Evaluate estimation errors over different current profiles and operating conditions. Develop a structured route towards data-driven SOH prediction using suitable battery datasets and carefully defined training and validation procedures.

5. **Develop the digital-twin framework**

   Establish consistent interfaces between the physical battery plant, sensor signals, estimated states, parameter updates, and supervisory control. Compare model predictions against measured or reference data wherever such data are available.

6. **Introduce degradation-aware charging optimization**

   Formulate an optimization objective that considers charging time, current limits, voltage limits, temperature, SOC, and degradation stress. The objective is to reduce unnecessary operating stress while respecting battery safety constraints.

7. **Prepare for hardware implementation**

   Identify the sensors, measurement interfaces, protection mechanisms, controller interfaces, and experimental procedures needed for a future hardware prototype. Hardware testing will require appropriate electrical protection and supervision.

1 Model overview and sign convention

The single-cell plant is represented by a two-RC Thevenin equivalent-circuit model (ECM), coupled to a lumped thermal model and a throughput-based ageing model. The electrical subsystem predicts terminal voltage and polarization dynamics; the SOC subsystem tracks stored charge; the thermal subsystem predicts cell temperature; and the ageing subsystem updates capacity and ohmic resistance as degradation accumulates.

The current sign convention used throughout the model is

$$i > 0 \quad \text{during discharge}, \qquad i < 0 \quad \text{during charge}$$

The state variables are the cell SOC, polarization voltages v_1 and v_2, cell temperature t_b, capacity-loss fraction d_Q, and resistance-growth fraction d_R. The notation uses lower-case circuit quantities for voltage, current, resistance, and capacitance.

2 Two-RC Thevenin equivalent circuit

2.1 Circuit representation

The ECM contains an open-circuit voltage source v_{ocv}, a series ohmic resistance r_0, and two polarization branches. Each polarization branch is represented by a resistor-capacitor pair. The polarization states account approximately for transient voltage relaxation that cannot be represented by a single ohmic resistance.

2.2 Derivation of the first polarization branch: r_1c_1

For the first RC branch, Kirchhoff's current law gives

$$i = i_{r_1} + i_{c_1}.$$

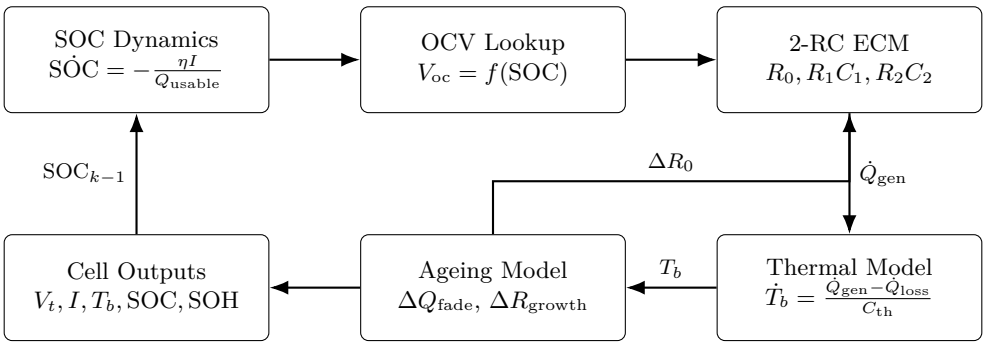


Figure 1: Functional decomposition of the single-cell plant.

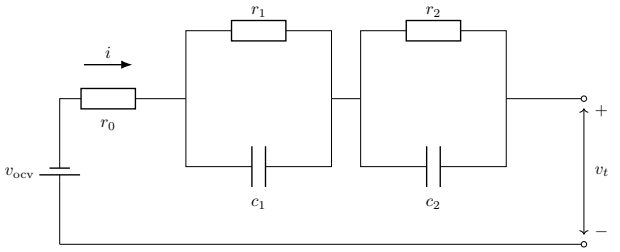


Figure 2: Two-RC Thevenin equivalent model of a lithium-ion battery cell.

Ohm's law for the resistor gives

$$i_{r_1} = \frac{v_1}{r_1},$$

and the capacitor current-voltage relationship is

$$i_{c_1} = c_1 \frac{\mathrm{d}v_1}{\mathrm{d}t}.$$

Substituting these expressions into Kirchhoff's current law,

$$i = \frac{v_1}{r_1} + c_1 \frac{\mathrm{d}v_1}{\mathrm{d}t}.$$

Rearranging for the derivative yields the state equation used in the Simulink model:

$$\boxed{\frac{\mathrm{d}v_1}{\mathrm{d}t} = \frac{i}{c_1} - \frac{v_1}{r_1 c_1}.}$$

The time constant of this branch is

$$\boxed{\tau_1 = r_1 c_1.}$$

For a constant current i and initial condition $v_1(0) = 0$, the solution is

$$\boxed{v_1(t) = i r_1 \left(1 - e^{-t/\tau_1}\right).}$$

2.3 Derivation of the second polarization branch: r_2c_2

The second RC branch is modelled independently using its own resistor r_2, capacitor c_2, and polarization voltage v_2. Applying Kirchhoff's current law,

$$i = i_{r_2} + i_{c_2}.$$

The branch resistor current is

$$i_{r_2} = \frac{v_2}{r_2},$$

while the capacitor current is

$$i_{c_2} = c_2 \frac{\mathrm{d}v_2}{\mathrm{d}t}.$$

Therefore,

$$i = \frac{v_2}{r_2} + c_2 \frac{\mathrm{d}v_2}{\mathrm{d}t}.$$

Solving for the state derivative gives the second polarization equation:

$$\boxed{\frac{\mathrm{d}v_2}{\mathrm{d}t} = \frac{i}{c_2} - \frac{v_2}{r_2 c_2}.}$$

Its time constant is

$$\boxed{\tau_2 = r_2 c_2.}$$

For constant current and $v_2(0) = 0$, the response is

$$\boxed{v_2(t) = i r_2 \left(1 - e^{-t/\tau_2}\right).}$$

The larger time constant represents a slower polarization-relaxation component than the r_1c_1 branch in this parameterization.

2.4 Combined state-space representation

Combining both RC branches produces

$$\boxed{\begin{bmatrix} \dot{v}_1 \\ \dot{v}_2 \end{bmatrix} = \begin{bmatrix} -\dfrac{1}{r_1 c_1} & 0 \\ 0 & -\dfrac{1}{r_2 c_2} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + \begin{bmatrix} \dfrac{1}{c_1} \\ \dfrac{1}{c_2} \end{bmatrix} i.}$$

Both states are initialized to zero in the present cell model:

$$\boxed{v_1(0) = 0, \qquad v_2(0) = 0.}$$

2.5 Terminal-voltage equation

The variable d_R represents the fractional resistance growth caused by battery ageing. It is a dimensionless state variable that quantifies the relative increase in the cell's ohmic resistance compared with its baseline value $r_{0,\text{base}}$.

The current ohmic resistance is modelled as

$$\boxed{r_0 = r_{0,\text{base}}(1 + d_R).}$$

When the battery is new, the resistance-growth state is initialized to zero:

$$\boxed{d_R(0) = 0.}$$

Consequently, the initial resistance is $r_0 = r_{0,\text{base}}$. As ageing accumulates, d_R increases and the effective ohmic resistance rises.

In the implemented ageing model, resistance growth is driven by charge throughput and operating stress:

$$\boxed{\frac{\mathrm{d}d_R}{\mathrm{d}t} = k_R f_{\text{stress}} \frac{|i|}{3600},}$$

where k_R is the resistance-growth coefficient and f_{stress} is the combined temperature- and SOC-dependent stress factor. This equation represents a simplified empirical ageing approximation rather than a fully calibrated electrochemical degradation law.

3 Open-circuit voltage model

$$v_{\text{ocv}} = f_{\text{ocv}}(\text{soc}).$$

The final lookup relationship is

$$\boxed{v_{\text{ocv}} = f_{\text{ocv}}(\text{soc}), \quad 0 \le \text{soc} \le 1.}$$

4 State-of-charge model

Let q denote the remaining charge in Ah and q_{avail} the available capacity in Ah. The definition of SOC is

$$\text{soc} = \frac{q}{q_{\text{avail}}}.$$

The parameter $\eta(i)$ represents the coulombic efficiency of the battery and accounts for the effective charge transferred during charging and discharging. It is a dimensionless quantity whose value depends on the direction of the battery current. The discharge efficiency, η_{dis}, is used when the current is non-negative, whereas the charging efficiency, η_{chg}, is used when the current is negative.

With discharge current defined as positive, the remaining charge decreases:

$$\frac{\mathrm{d}q}{\mathrm{d}t} = -\frac{\eta(i)i}{3600}.$$

If capacity is locally constant over the SOC integration step,

$$\frac{\mathrm{d}\,\text{soc}}{\mathrm{d}t} = \frac{1}{q_{\text{avail}}} \frac{\mathrm{d}q}{\mathrm{d}t}.$$

Thus the SOC differential equation is

$$\boxed{\dot{\text{soc}} = -\frac{\eta(i)i}{3600 q_{\text{avail}}}.}$$

For signed-current efficiency,

$$\eta(i) = \begin{cases} \eta_{\text{dis}}, & i \ge 0, \\ \eta_{\text{chg}}, & i < 0. \end{cases}$$

The implemented state is limited to the physical interval:

$$\boxed{\text{soc}(t) = \text{sat}_{[0,1]} \left(\text{soc}(0) - \int_0^t \frac{\eta(i(\lambda))i(\lambda)}{3600 q_{\text{avail}}(\lambda)} \,\mathrm{d}\lambda \right)}$$

5 Thermal model

5.1 Thermal energy balance

Let t_b be cell temperature, t_a ambient temperature, c_{th} lumped thermal capacitance, and ha the lumped heat-transfer coefficient times area. The rate of thermal energy stored in the cell is

$$\dot{e}_{\text{th}} = c_{\text{th}} \dot{t}_b.$$

The heat rejected to the ambient is approximated by Newton's law of cooling:

$$p_{\text{loss}} = ha(t_b - t_a).$$

The current implementation calculates generated heat as

$$p_{\text{gen}} = i^2 r_0 + \frac{v_1^2}{r_1} + \frac{v_2^2}{r_2}.$$

Applying conservation of energy,

$$c_{\text{th}} \dot{t}_b = p_{\text{gen}} - p_{\text{loss}}.$$

Therefore, the final thermal equation used in the model is

$$\boxed{\dot{t}_b = \frac{i^2 r_0 + \frac{v_1^2}{r_1} + \frac{v_2^2}{r_2}}{c_{\text{th}}} - \frac{ha(t_b - t_a)}{c_{\text{th}}}.}$$

The generated heat includes the ohmic loss and the two RC-resistor loss terms used in the project's implementation. This is a simplified thermal model; reversible entropic heat and detailed heat-transfer paths are not included.

5.2 Thermal circuit analogy

Temperature difference corresponds to voltage, heat flow corresponds to current, thermal resistance corresponds to electrical resistance, and thermal capacitance corresponds to electrical capacitance. Define

$$\theta = t_b - t_a, \qquad r_{\text{th}} = \frac{1}{ha}.$$

The thermal network is represented by a heat-flow source feeding a thermal capacitance and a thermal resistance to the ambient reference.

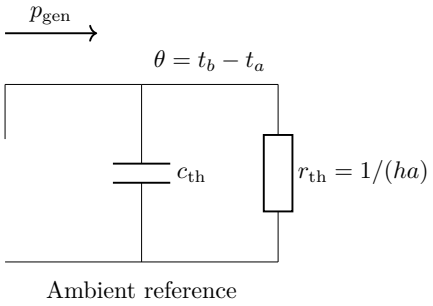


Figure 3: Lumped thermal network analogy.

The thermal time constant is

$$\boxed{\tau_{\text{th}} = \frac{c_{\text{th}}}{ha}.}$$

6 Ageing and degradation model

6.1 Absolute charge throughput

The absolute throughput state counts charge transferred in either direction. Since current is in amperes and time is in seconds, division by 3600 converts coulombs to Ah:

$$\boxed{\dot{a}_{\text{th}} = \frac{|i|}{3600}.}$$

With a sample period T_s, the discrete update is

$$\boxed{a_{\text{th}}[k] = a_{\text{th}}[k-1] + \frac{|i[k]|T_s}{3600}.}$$

6.2 Stress-factor calculation

Temperature and SOC stress factors are obtained from lookup functions $f_T(t_b)$ and $f_{\text{soc}}(\text{soc})$. The combined stress factor is

$$\boxed{f_{\text{stress}} = f_T(t_b) f_{\text{soc}}(\text{soc}).}$$

These factors are illustrative project assumptions, not a chemistry-specific ageing law established by the manufacturer.

6.3 Capacity fade

Let d_Q be a dimensionless capacity-loss fraction and k_Q a capacity fade coefficient with units Ah^{-1}. The ageing rate is assumed proportional to absolute charge throughput and stress:

$$\dot{d}_Q = k_Q f_{\text{stress}} \frac{|i|}{3600}.$$

Integrating from the initial condition $d_Q(0) = 0$ gives

$$d_Q(t) = k_Q \int_0^t f_{\text{stress}}(\lambda) \frac{|i(\lambda)|}{3600}\, \mathrm{d}\lambda.$$

The physical SOH and available capacity are then

$$\boxed{\text{soh}_{\text{true}} = \text{sat}_{[0,1]}(1 - d_Q),}$$

$$\boxed{q_{\text{avail}} = q_{\text{nom}} \text{soh}_{\text{true}}.}$$

The current profile uses $q_{\text{nom}} = 3.45\,\text{Ah}$ and $k_Q = 2 \times 10^{-5}\,\text{Ah}^{-1}$. The coefficient is provisional.

6.4 Resistance growth

Let d_R be a dimensionless resistance-growth fraction. Its evolution is modelled as

$$\dot{d}_R = k_R f_{\text{stress}} \frac{|i|}{3600}.$$

For $d_R(0) = 0$,

$$d_R(t) = k_R \int_0^t f_{\text{stress}}(\lambda) \frac{|i(\lambda)|}{3600}\, \mathrm{d}\lambda.$$

The current resistance is

$$\boxed{r_0 = r_{0,\text{base}}(1 + d_R).}$$

The profile uses $r_{0,\text{base}} = 0.045\,\Omega$ and $k_R = 3 \times 10^{-5}\,\text{Ah}^{-1}$. This throughput law does not represent calendar ageing or detailed electrochemical degradation.

6.5 Ageing signal-flow diagram

6.6 BMS SOH estimate used in the current model

The BMS estimator uses a simplified temperature-weighted throughput accumulator, distinct from the physical ageing states:

$$\dot{a}_T = f_T(t_m) \frac{|i_f|}{3600}.$$

The final estimator equation is

$$\boxed{\text{soh}_{\text{est}} = \text{sat}_{[0,1]}\left(\text{soh}_0 - k_Q a_T\right).}$$

Here i_f is filtered current and t_m is measured temperature. The current estimator does not include a voltage-stress term. It is a model-based indicator and has not been validated as a field SOH measurement.

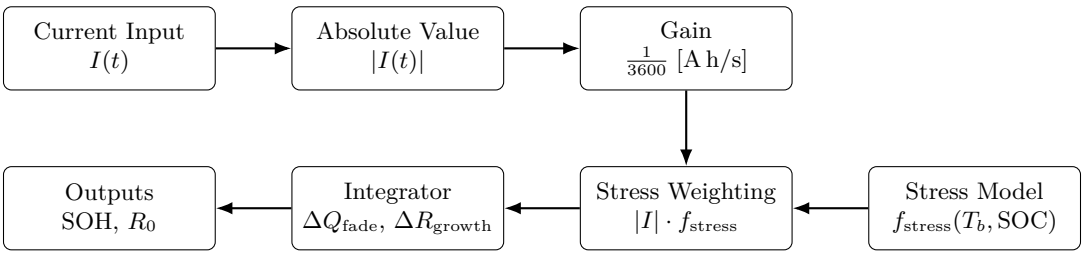


Figure 4: Conceptual signal flow for the throughput-based ageing calculation.

7 Single-cell parameter set

Parameter	Current value	Description
q_{nom}	3.45 Ah	Nominal capacity in selected profile
v_{nom}	3.60 V	Nominal voltage in selected profile
v_{max}	4.20 V	Upper voltage limit in profile
v_{min}	2.50 V	Lower voltage limit in profile
$i_{\text{chg,max}}$	1.70 A	Maximum charge-current limit in profile
$i_{\text{dis,max}}$	10.0 A	Maximum discharge-current limit in profile
soc(0)	0.95	Initial SOC
$t_b(0)$	45 °C	Initial temperature in profile
r_0	0.045 Ω	Base ohmic resistance; provisional
r_1	0.020 Ω	First polarization resistance; provisional
c_1	2200 F	First polarization capacitance; provisional
r_2	0.012 Ω	Second polarization resistance; provisional
c_2	15 000 F	Second polarization capacitance; provisional
η_{chg}	0.995	Charge efficiency
η_{dis}	0.995	Discharge efficiency
c_{th}	500 J/K	Lumped thermal capacitance; provisional
ha	1.2 W/K	Heat-transfer coefficient-area product; provisional
t_a	42 °C	Ambient temperature used in baseline test
k_Q	2×10^{-5} Ah^{-1}	Capacity fade coefficient; provisional
k_R	3×10^{-5} Ah^{-1}	Resistance growth coefficient; provisional

The selected profile name references Molicel INR-18650-M35A. The values for r_0, r_1, c_1, r_2, c_2, the OCV table, thermal parameters, and ageing coefficients are model assumptions and must not be described as calibrated manufacturer parameters without supporting identification data.

8 Topology and pack abstraction

The current Simulink implementation uses eight copies of the single-cell subsystem connected in series. Let $v_{t,j}$, i_j, t_j, soc_j, soh_j, $q_{\text{avail},j}$, and $r_{0,j}$ denote the states and outputs of cell j, where $j = 1, \ldots, 8$.

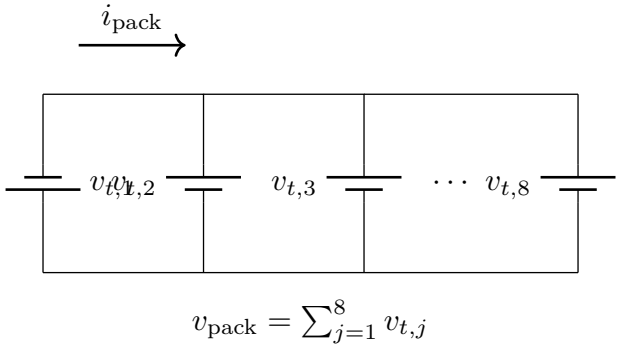


Figure 5: Eight single-cell models connected in series.

9 Series current and voltage derivation

For ideal series-connected cells, Kirchhoff's current law implies that the same current flows through every cell:

$$i_1 = i_2 = \cdots = i_8 = i_{\text{pack}}.$$

Kirchhoff's voltage law gives the pack voltage as the sum of cell terminal voltages:

$$v_{\text{pack}} = v_{t,1} + v_{t,2} + \cdots + v_{t,8}.$$

Hence the final pack equations are

$$\boxed{i_{\text{pack}} = i_j, \quad j = 1, \ldots, 8,}$$

$$\boxed{v_{\text{pack}} = \sum_{j=1}^{8} v_{t,j}.}$$

If all cells are identical and have the same state,

$$\boxed{v_{\text{pack}} = 8 v_t.}$$

The equality is an ideal identical-cell result; cell mismatch can make the individual terminal voltages differ.

10 Pack resistance

The incremental ohmic resistances of series-connected cells add:

$$r_{0,\mathrm{pack}} = r_{0,1} + r_{0,2} + \cdots + r_{0,8}.$$

Therefore,

$$\boxed{r_{0,\mathrm{pack}} = \sum_{j=1}^{8} r_{0,j},}$$

$$\boxed{r_{0,\mathrm{pack}} = 8r_0 \quad \text{for identical cells.}}$$

This expression represents the series sum of the cells' ohmic resistance states. It does not include busbar, fuse, contactor, connector, or cable resistance unless those elements are added separately.

11 Pack SOC and SOH aggregation

In the current pack subsystem, SOC is aggregated using the arithmetic mean:

$$\boxed{\mathrm{soc}_{\mathrm{pack}} = \frac{1}{8}\sum_{j=1}^{8} \mathrm{soc}_j.}$$

The mean is a useful descriptive pack indicator when cell capacities are similar, but it does not guarantee that every cell remains within its safe operating range. A conservative pack SOH is calculated using the minimum cell SOH:

$$\boxed{\mathrm{soh}_{\mathrm{pack}} = \min_{j=1,\ldots,8} \mathrm{soh}_j.}$$

This minimum-cell rule highlights the weakest modelled cell. A future pack estimator may use capacity-weighted aggregation or a state observer that explicitly models cell imbalance.

12 Available pack capacity

Series connection increases voltage but does not multiply Ah capacity. For an ideal series string, the same Ah passes through every cell. A conservative available-capacity output is therefore

$$\boxed{q_{\mathrm{pack,avail}} = \min_{j=1,\ldots,8} q_{\mathrm{avail},j}.}$$

For eight identical cells,

$$\boxed{q_{\mathrm{pack,avail}} = q_{\mathrm{cell,avail}},}$$

$$\boxed{e_{\mathrm{nom,pack}} \approx 8 v_{\mathrm{nom}} q_{\mathrm{nom}}.}$$

Here nominal energy in Wh is approximated by nominal voltage times capacity; actual energy depends on the voltage trajectory during discharge.

13 Pack temperature

The pack subsystem reports the maximum cell temperature:

$$\boxed{t_{\mathrm{pack,max}} = \max_{j=1,\ldots,8} t_j.}$$

This is a conservative thermal monitoring indicator. It is not the average pack temperature and does not model thermal coupling between neighbouring cells.

14 Pack charge throughput

Because the current is identical in all series cells, every cell experiences the same absolute charge throughput if the initial throughput states and current histories are identical:

$$\dot{a}_{\mathrm{th},j} = \frac{|i_{\mathrm{pack}}|}{3600}.$$

The pack throughput should be reported from one representative cell:

$$\boxed{a_{\mathrm{th,pack}} = a_{\mathrm{th},j} \quad \text{for any } j \text{ in an ideal series string.}}$$

Summing all eight cell throughput states gives eight times the charge throughput and should not be labelled simply as pack Ah throughput. If such a sum is retained for diagnostic purposes, label it explicitly as total cell-throughput accumulation.

15 Pack subsystem equations used

For the present eight-cell series abstraction, the pack-level output definitions are summarised as

$$\boxed{\begin{aligned}
v_{\mathrm{pack}} &= \sum_{j=1}^{8} v_{t,j}, \\
i_{\mathrm{pack}} &= i_1 = \cdots = i_8, \\
t_{\mathrm{pack,max}} &= \max_j t_j, \\
\mathrm{soc}_{\mathrm{pack}} &= \frac{1}{8}\sum_{j=1}^{8} \mathrm{soc}_j, \\
\mathrm{soh}_{\mathrm{pack}} &= \min_j \mathrm{soh}_j, \\
q_{\mathrm{pack,avail}} &= \min_j q_{\mathrm{avail},j}, \\
r_{0,\mathrm{pack}} &= \sum_{j=1}^{8} r_{0,j}, \\
a_{\mathrm{th,pack}} &= a_{\mathrm{th},j}.
\end{aligned}}$$

These equations describe the aggregation rules intended for the current Simulink subsystem. Each output should be checked against its actual Mux, Sum, MinMax, and signal-routing blocks to ensure that the diagram implements the documented expression.

16 Existing BMS Architecture

The battery management system (BMS) is implemented as a supervisory subsystem between the requested battery current and the physical battery plant. The subsystem receives measured terminal voltage, current, and temperature, together with the requested current command. It processes these signals and generates estimated states, protection flags, and a safe current command.

The principal functions implemented in the current model are:

1. Filtering of measured voltage, current, and temperature.
2. SOC estimation using current integration.
3. Simplified SOH estimation using temperature-weighted charge throughput.
4. Overvoltage and undervoltage detection.
5. Charge and discharge overcurrent detection.
6. Overtemperature and undertemperature detection.
7. Fault-state generation and supervisory current limiting.
8. Emergency shutdown and safe-current command generation.

The BMS is connected to the battery plant through a discrete delay element to avoid a direct algebraic loop in the closed-loop Simulink model. During emergency shutdown, the intended safe-current command is zero.

17 Baseline Single-Cell Validation

A baseline simulation was conducted using the selected Molicel INR-18650-M35A profile, a constant discharge current of 0.5 A, and a simulation duration of 300 s. The initial SOC was 0.95, and the ambient temperature was 42 °C.

The principal results are summarised in Table 1.

Table 1: Baseline single-cell simulation results.

Metric	Observed result
Simulation duration	300 s
Applied discharge current	0.5 A
Terminal voltage range	3.9233 V to 3.9800 V
Battery temperature range	43.466 °C to 45 °C
Initial SOC	0.950000
Final true SOC	0.940126
Final estimated SOC	0.940036
SOC mean absolute error	0.00003297 (fraction)
Initial true SOH	1.000000
Final true SOH	0.99999826
Final estimated SOH	0.99999868

The baseline simulation completed successfully, and the SOC and SOH signals remained within their configured bounds. The SOC estimation error was small for this particular current profile and operating condition. However, these results demonstrate internal consistency for the tested simulation rather than validated accuracy against a physical battery. Additional profiles and experimental or reference data are required for broader validation.

18 BMS Protection Testing

A protection test suite has been developed to examine normal operation, overtemperature, undertemperature, discharge overcurrent, and charge overcurrent conditions. The tests examine fault flags, fault codes, BMS states, and the safe-current output.

The normal-operation and temperature-fault tests have produced expected responses in the current test configuration. The discharge overcurrent test has also revealed an incorrect charge-overcurrent indication. Therefore, the overcurrent logic requires correction before the protection subsystem can be considered fully validated.

The next verification steps are:

1. Correct the charge and discharge overcurrent conditions.
2. Test positive and negative current commands separately.
3. Verify fault flags and fault codes against each injected fault.
4. Check the transitions between normal, limited, fault, and emergency states.
5. Confirm that emergency shutdown forces the battery current command to zero.
6. Repeat the complete test suite after the corrections.

19 Pack-Model Reconciliation

The existing pack subsystem consists of eight copies of the developed single-cell model connected in series. The professor-provided reference model must be examined in detail to determine its module definition, cell scaling, electrical parameters, and state aggregation conventions.

The implementation will be reconciled with the reference model before additional pack-level control features are introduced. Verification will include the common series current, summed terminal voltage, equivalent resistance, available capacity, temperature monitoring, and SOC and SOH reporting.

19.1 Performance Metrics for SOC Estimation

The accuracy of the SOC estimator is evaluated by comparing the estimated SOC with the reference SOC over N samples.

19.1.1 Mean Absolute Error (MAE)

The Mean Absolute Error measures the average absolute difference between the estimated and reference SOC values:

$$\boxed{\mathrm{MAE} = \frac{1}{N}\sum_{k=1}^{N} |\mathrm{SOC}_{\mathrm{est},k} - \mathrm{SOC}_{\mathrm{ref},k}|}$$

19.1.2 Mean Squared Error (MSE)

The Mean Squared Error calculates the average of the squared differences between the estimated and reference SOC values:

$$\boxed{\mathrm{MSE} = \frac{1}{N}\sum_{k=1}^{N} (\mathrm{SOC}_{\mathrm{est},k} - \mathrm{SOC}_{\mathrm{ref},k})^2}$$

Squaring the errors gives greater weight to larger estimation errors. The MSE is expressed in squared SOC units.

19.1.3 Root Mean Squared Error (RMSE)

The Root Mean Squared Error is the square root of the MSE:

$$\boxed{\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{k=1}^{N} (\mathrm{SOC}_{\mathrm{est},k} - \mathrm{SOC}_{\mathrm{ref},k})^2}}$$

The RMSE is expressed in the same units as SOC and provides a measure of the magnitude of estimation errors.

20 Simulink model Integration

The digital-twin framework will connect the physical battery model, sensor measurements, state estimators, and supervisory controller through consistent signal interfaces. The estimated states and model outputs will be compared against corresponding plant or reference signals.

The intended workflow is:

1. Acquire or generate voltage, current, and temperature measurements.
2. Estimate battery SOC and SOH from the available signals.
3. Compare measured outputs with the digital-twin predictions.
4. Quantify prediction errors and identify parameter mismatch.
5. Investigate suitable model-parameter updates.
6. Feed validated estimates and operating limits into the supervisory controller.

Online parameter adaptation will be introduced only after the baseline model and estimation interfaces have been verified.

21 AI/ML-Based SOH Prediction

A data-driven SOH prediction stage is planned as an extension of the physics-based model. Suitable battery ageing datasets will be selected according to the availability of current, voltage, temperature, cycle-history, and capacity-reference information.

Candidate input features may include charge throughput, voltage statistics, temperature statistics, current-profile characteristics, and cycle-related features. Candidate regression models will be evaluated using separate training, validation, and test data.

The final model will be selected using prediction error, generalisation performance, computational requirements, and suitability for eventual deployment. Data leakage between training and test sets will be avoided. The selected model will be integrated with the BMS only after its performance has been evaluated on held-out data.

22 Degradation-Aware Charging Optimization

A charging optimization strategy will be developed to balance charging time and degradation-related operating stress while satisfying battery safety constraints.

A general objective may be expressed as

$$\min_{\{i(k)\}} J = w_t J_{\mathrm{time}} + w_d J_{\mathrm{degradation}} + w_T J_{\mathrm{thermal}},$$

where J_{time} represents charging-time cost, $J_{\mathrm{degradation}}$ represents a degradation-related cost, J_{thermal} represents thermal stress, and the w coefficients are non-negative weighting parameters.

The optimization will respect constraints such as

$$v_{\min} \le v_t(k) \le v_{\max},$$

$$-i_{\mathrm{chg,max}} \le i(k) \le i_{\mathrm{dis,max}},$$

$$t_{\min} \le t_b(k) \le t_{\max}, \qquad 0 \le \mathrm{SOC}(k) \le 1.$$

The objective, weighting coefficients, and constraints will be refined according to the intended charging scenario and available battery data. The optimization strategy will be compared with a baseline charging policy using charging time, temperature response, constraint violations, and degradation-related model indicators.

23 Preparation for Hardware Implementation

The final stage of the planned work is to prepare the simulation framework for potential hardware implementation. This stage will identify suitable voltage, current, and temperature sensors, data acquisition interfaces, controller hardware, and independent protection mechanisms.

The hardware design will require appropriate isolation, current limiting, overvoltage protection, thermal supervision, and emergency disconnection. Any experimental testing will follow a documented procedure with suitable laboratory supervision and electrical safety measures.